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SEMICONDUCTOR STRUCTURE AND METHOD FOR FABRICATING THE SAME

Abstract

A method of fabricating a semiconductor structure includes: providing a first wafer; providing a second wafer having a first surface and a second surface opposite to the first surface; contacting the first surface of the second wafer with the first wafer; and forming a plurality of scribe lines on the second surface of the second wafer, wherein the formation of the plurality of scribe lines includes removing portions of the second wafer from the second surface towards the first surface to form a third surface between the first surface and the second surface, and the plurality of scribe lines protrudes from the third surface of the second wafer.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is continuation of U.S. application Ser. No. 18/359,889, filed on Jul. 27, 2023, which is a continuation of U.S. application Ser. No. 16/932,622, filed on Jul. 17, 2020, which is a division of U.S. application Ser. No. 15/168,848, filed on May 31, 2016, which application is hereby incorporated herein by reference.

BACKGROUND

[0002] Semiconductor devices are used in a variety of electronic applications, such as personal computers, cell phones, and digital cameras. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers over a semiconductor substrate, and patterning the various layers using lithography to form circuit components and elements thereon.

[0003] The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. Individual dies on a wafer may be formed using semiconductor processing techniques. After device formation, the dies may be cut from the wafer by a saw, laser or the like. The dies may be spaced apart by a dicing street, scribe line or the like to account for the width of the cutting apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is diagram illustrating a top view of a wafer level semiconductor structure with a plurality of scribe lines disposed on a surface of the wafer level semiconductor structure according to some embodiments.

[0006] FIG. 2 is a diagram illustrating a cross-sectional view of a portion of the wafer level semiconductor structure taken along the line A-A' shown in FIG. 1 according to some embodiments.

[0007] FIG. 3 is a diagram illustrating a top view of a portion of the wafer level semiconductor structure of FIG. 1 according to some embodiments.

[0008] FIG. 4 is a diagram illustrating a cross-sectional view of a portion of the wafer level semiconductor structure taken along the line B-B' shown in FIG. 1 according to some embodiments.

[0009] FIG. 5 to FIG. 13 are cross-sectional views of a CMOS-MEMS wafer fabricated at various stages, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0010] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0011] Embodiments of the present disclosure are discussed in detail below. It should be appreciated, however, that the present disclosure provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative and do not limit the scope of the disclosure.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “lower,” “left,” “right” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. It will be understood that when an element is referred to as being “connected to” or “coupled to” another element, it may be directly connected to or coupled to the other element, or intervening elements may be present.

[0013] Micro-electro mechanical system (MEMS) devices have been developed and used in electronic equipment. In MEMS device fabrication, semiconductive materials are used to form mechanical and electrical features. A MEMS device may include a number of elements (e.g., stationary or movable elements) for achieving electro-mechanical functionality. MEMS applications include motion sensors, pressure sensors, printer nozzles, or the like. Complementary Metal-Oxide-Semiconductor (CMOS) devices may be analog circuits such as CMOS image sensors (CIS), analog to digital converters used in microprocessors, microcontrollers, Random Access Memories (RAM). A MEMS device and a CMOS device may be integrated into a die. During the fabrication process, a MEMS wafer may be bonded with a CMOS wafer before the dicing process. The MEMS-CMOS wafer may be thinned into die level when the semiconductor industry continues to reduce the feature size of a MEMS-CMOS die. Large warpage may damage or delaminate the structure in the MEMS-CMOS wafer during the transportation of the MEMS-CMOS wafer.

[0014] To solve the above mentioned problem, a scribe line structure capable of increasing a wafer tension is proposed. The scribe line structure is disposed on a surface or a side of a wafer for alleviating the warpage or deformation of the wafer during transportation. The wafer may be a MEMS wafer, a CMOS wafer, or a MEMS-CMOS wafer. For brevity, the feature of the proposed scribe line structure is carried out by a MEMS-CMOS wafer. The scribe line structure comprises a plurality of scribe lines positioned between dies of the MEMS-CMOS wafer to permit the use of a cutting apparatus to singulate the dies without cutting the dies themselves. Each scribe line has a width to accommodate the cutting apparatus and a buffer region to account for variances in cutting or the cutting apparatus.

[0015] FIG. 1 is diagram illustrating a top view of a wafer level semiconductor structure **100** with a plurality of scribe lines **102** disposed on a surface **104** of the semiconductor structure **100** according to some embodiments. The semiconductor structure **100** may be a MEMS-CMOS wafer.

The MEMS-CMOS wafer comprises a MEMS wafer bonding with a CMOS wafer. According to some embodiments, the plurality of scribe lines **102** is disposed on a backside of the semiconductor structure **100**, i.e., the surface **104** of the MEMS wafer. This is not a limitation of the embodiments. The plurality of scribe lines **102** may be disposed on a front side of the semiconductor structure **100**, i.e., the surface of the CMOS wafer. The plurality of scribe lines **102** protrudes from the surface **104** of the semiconductor structure **100**. The semiconductor structure **100** comprises an array of dies (or chips) **106** separated from each other by two sets of intersecting scribe lines **102**. A first set of scribe lines **102** may be a plurality of straight lines extend along a first direction and a second set of scribe lines **102** may be a plurality of straight lines extend along a second direction. The first direction may be perpendicular to the second direction. However, this is not a limitation of the embodiments. The first direction and the second direction may form any angle on the intersections.

[0016] The semiconductor structure **100** is further configured to have a wafer edge line **108** on the wafer edge of the semiconductor structure **100**. The wafer edge line **108** also protrudes from the surface **104** of the semiconductor structure **100**. The wafer edge line **108** surrounds the plurality of scribe lines **102**. According to some embodiments, each of the scribe lines **102** has one end contacting an inner point of the wafer edge line **108** and the other end contacting a corresponding inner point of the wafer edge line **108**. The line width of the wafer edge line **108** is larger than the line width of a scribe line. For example, the line width of the wafer edge line **108** may be about 5 mm and the line width of a scribe line **102** may be about 60 μm , as shown in FIG. 1.

[0017] According to some embodiments, the plurality of scribe lines **102** and the wafer edge line **108** have the similar height measured from the surface **104** of the semiconductor structure **100**. In other words, the plurality of scribe lines **102** and the wafer edge line **108** have the similar top level. The plurality of scribe lines **102** and the wafer edge line **108** are arranged to protrude from the surface **104** of the semiconductor structure **100** to increase the effective thickness of the semiconductor structure **100** before being diced. When the effective thickness of the semiconductor structure **100** increases, the wafer tension of the semiconductor structure **100** also increases. Accordingly, the warpage or deformation of the semiconductor structure **100** can be alleviated when the handling system handles the semiconductor structure **100** during transportation.

[0018] The scribe lines **102** may have different contact areas to contact the surface **104** at the bottom sides. FIG. 2 is a diagram illustrating a cross-sectional view of a portion **200** of the semiconductor structure **100** taken along the line A-A' shown in FIG. 1 according to some embodiments. FIG. 2 merely shows a portion **200** of the semiconductor structure **100**, and this is not a limitation of the present disclosure. The semiconductor structure **100** comprises two wafers, in which the first wafer is a MEMS wafer **202** and the second wafer is a CMOS wafer **204**. The MEMS wafer **202** is configured to form a structure corresponding to an electro-mechanical function. The CMOS wafer **204** is configured to form an integrated circuit, such as an application-specific integrated circuit (ASIC). The MEMS wafer **202** has a first surface **206** bonding to a predetermined bonding area of the CMOS wafer **204** such that the MEMS structures of the array of dies **106** on the MEMS wafer **202** are aligned to the integrated circuits of the array of dies **106** on the CMOS wafer **204**. The MEMS wafer **202** has a second surface **208** being the top surface of the semiconductor structure **100**. The level of the surface **104** is sandwiched between the levels of the surface **206** and the surface **208**.

[0019] According to some embodiments, the portion **200** at least comprises three regions, in which the first region is a first scribe line region **210**, the second region is a die region **212**, and the third region is a second scribe line region **214**. The die region **212** is disposed between the first scribe line region **210** and the second scribe line region **214**. For illustration purpose, the structure of the first scribe line region **210** is different from the structure of the second scribe line region **214**. However, this is not a limitation of the present disclosure. All of the scribe lines **102** of the semiconductor structure **100** may be arranged to have the structure similar to the structure of the

first scribe line region **210**, or all of the scribe lines **102** of the semiconductor structure **100** may be arranged to have the structure similar to the structure of the second scribe line region **214**, or some of the scribe lines **102** of the semiconductor structure **100** are arranged to have the structure similar to the structure of the first scribe line region **210** and the other scribe lines **102** of the semiconductor structure **100** are arranged to have the structure similar to the structure of the second scribe line region **214**.

[0020] According to the portion **200** as shown in FIG. 2, a die **106** is formed in the die region **212**. The first scribe line region **210** and the second scribe line region **214** are to be cut by a cutting tool to singulate the die **106**. The first scribe line region **210** is arranged to form a first scribe line **218**. The second scribe line region **214** is arranged to form a second scribe line **220**. The first scribe line **218** and the second scribe line **220** are arranged to protrude from the surface **104** to increase the wafer tension of the semiconductor structure **100**.

[0021] According to some embodiments, the first scribe line **218** and the second scribe line **220** are formed by a semiconductor layer **222** of the MEMS wafer **202**. The semiconductor layer **222** may be the substrate of the MEMS wafer **202**. In the first scribe line region **210**, the first scribe line **218** has a surface contacting with a first surface (i.e. **104**) of an insulating line **224**. A second surface of the insulating line **224** is contacted with a semiconductor layer **226**. In the second scribe line region **214**, the second scribe line **220** has a surface contacting with a first surface (i.e. **104**) of a semiconductor line **228**. A second surface of the semiconductor line **228** is contacted with the CMOS wafer **204** at the surface **206**. The material of the insulating line **224** is the sacrificial material. For example, the material of the insulating line **224** may be silicon dioxide (SiO_2), and this is not a limitation of the present disclosure. The material of the semiconductor line **228** is the structure material of the MEMS wafer **202**. For example, the material of the semiconductor line **228** may be polycrystalline silicon, also called polysilicon or poly-Si, and this is not a limitation of the present disclosure. The materials of the scribe lines **218**, **220**, and the semiconductor layer **226** are the structure material of the MEMS wafer **202**. For example, the material of the scribe lines **218**, **220**, and the semiconductor layer **226** may be silicon, and this is not a limitation of the present disclosure.

[0022] For illustration purpose, it is assumed that the first scribe line **218** and the second scribe line **220** have the similar line width $W1$. The line width of the insulating line **224** is $W2$. The thickness between the surface **206** and the surface **104** is $H1$. The thickness of the semiconductor layer **222** (i.e. the height between the surface **208** and the surface **104**) is $H2$. The thickness of the CMOS wafer **204** is $H3$. The thickness of the die **106** is $H4$. The thickness of the semiconductor structure **100** is $H5$. According to some embodiments, in the first scribe line region **210**, the line width $W2$ of the insulating line **224** is smaller than the line width $W1$ of the first scribe line **218**. In the second scribe line region **214**, the line width of the semiconductor line **228** is substantially equal to the line width $W1$ of the second scribe line **220**. The line width $W1$ of the first scribe line **218** is arranged to be larger than the thickness $H1$. For example, the line width $W1$ of the first scribe line **218** may be at least two times larger than the thickness $H1$. For another example, the thickness $H1$ may be about 10 μm , the line width $W1$ may be about 60 μm , and the line width $W2$ may be about 20 μm . According to some embodiments, the thickness $H2$ of the semiconductor layer **222** may be about 290 μm . The thickness $H3$ of the CMOS wafer **204** may be about 240 μm . The thickness $H4$ of the die **106** may be about 250 μm . The thickness $H5$ of the semiconductor structure **100** may be about 540 μm .

[0023] According to some embodiments, the insulating line **224** is arranged to make the first scribe line **218** to fix on the surface **104** while the semiconductor line **228** is arranged to make the second scribe line **220** to fix on the surface **104**. When all of the scribe lines **102** are firmly fixed on the surface **104** of the semiconductor structure **100**, the effective thickness $H5$ of the semiconductor structure **100** is increased. When the effective thickness $H5$ of the semiconductor structure **100** is increased, the wafer tension of the semiconductor structure **100** also increases such that the

warpage or deformation of the semiconductor structure **100** can be reduced when the handling system handles the semiconductor structure **100** during transportation. In other words, by using the proposed scribe lines **102** to increase the effective thickness H5 of the semiconductor structure **100**, the array of dies **106** can be designed to have relatively small thickness H4 (e.g. 250 μm) while the handling system still can handle the semiconductor structure **100** without being delaminated.

[0024] FIG. **3** is a diagram illustrating a top view of a portion **300** of the semiconductor structure **100** according to some embodiments. The portion **300** only shows the area with the structure corresponding to the first scribe line region **210**. The insulating line **224** is disposed on the surface **104**. The first scribe line **218** is disposed on the insulating line **224**. Thus, the insulating line **224** is arranged to attach the first scribe line **218** to the surface **104**. The portion **302** of the first scribe line **218** is suspended over the surface **104**. The dies are surrounded by the insulating line **224** and the first scribe line **218**. The line width of the insulating line **224** is W2. The line width of the first scribe line **218** is W1.

[0025] According to some embodiments, the wafer edge line **108** is disposed on a semiconductor line in the wafer edge region of the MEMS wafer **202** as shown in FIG. **4**. FIG. **4** is a diagram illustrating a cross-sectional view of a portion **400** of the semiconductor structure **100** taken along the line B-B' shown in FIG. **1** according to some embodiments. FIG. **4** merely shows the portion **400** of the semiconductor structure **100**, and this is not a limitation of the present disclosure. For illustration purpose, some numerals in FIG. **2** are also labelled in FIG. **4**. According to some embodiments, the portion **400** at least comprises three regions, in which the first region is second scribe line region **214** as shown in FIG. **2**, the second region is a die region **402**, and the third region is a wafer edge region **404**. The die region **402** is disposed between the second scribe line region **214** and the wafer edge region **404**. For illustration purpose, the structure of the second scribe line region **214** in FIG. **4** is not a limitation of the present disclosure. The structure of the second scribe line region **214** may be the structure of the first scribe line region **210** in FIG. **4**.

[0026] According to the portion **400** as shown in FIG. **4**, a die **106** is formed in the die region **402**. The second scribe line region **214** and the wafer edge region **404** are to be cut by a cutting tool to singulate the die **106**. The second scribe line region **214** is arranged to form the second scribe line **220**. The wafer edge region **404** is arranged to form the wafer edge line **108**. The second scribe line **220** and the wafer edge line **108** are arranged to protrude from the surface **104** to increase the wafer tension of the semiconductor structure **100**.

[0027] According to some embodiments, the wafer edge line **404** is formed by the semiconductor layer **222** of the MEMS wafer **202**. In the wafer edge region **404**, the wafer edge line **108** has a surface contacting with a first surface (i.e. **104**) of a semiconductor line **408**. A second surface of the semiconductor line **408** is contacted with the CMOS wafer **204** at the surface **206**. The material of the semiconductor line **408** may be polycrystalline silicon. The material of the wafer edge line **108** may be silicon.

[0028] For illustration purpose, it is assumed that the line width of the wafer edge line **108** is W3. According to some embodiments, the line width of the semiconductor line **408** is substantially equal to the line width of the wafer edge line **108**, i.e. W3. The line width W3 of the wafer edge line **108** is larger than the line width W1 of the second scribe line **220**. For example, the line width W1 may be about 60 μm , and the line width W3 may be about 5 mm.

[0029] According to some embodiments, the semiconductor line **408** is arranged to make the wafer edge line **108** to fix on the surface **104**. The wafer edge line **108** surrounds the plurality of scribe lines **102**, and the two ends of each of the scribe lines **102** contact with the inner side of the wafer edge line **108** as shown in FIG. **1**.

[0030] Accordingly, when all of the scribe lines **102** and the wafer edge line **108** are firmly fixed on the surface **104** of the semiconductor structure **100**, not only the effective thickness H5 of the semiconductor structure **100** is increased, the wafer tension of the semiconductor structure **100** is further increased such that the warpage or deformation of the semiconductor structure **100** can be

reduced when the handling system handles the semiconductor structure **100** during transportation. In other words, by using the proposed scribe lines **102** and the wafer edge line **108** to increase the effective thickness **H5** and the wafer tension of the semiconductor structure **100**, the array of dies **106** can be designed to have relatively small thickness **H4** (e.g. 250 μm) while the handling system still can handle the semiconductor structure **100** without being delaminated.

[0031] For brevity, the formation of the portions **200** and **300** of the semiconductor structure **100** are described with reference to FIG. 5 to FIG. 13 in detail below. This is not a limitation of the present disclosure. FIG. 5 to FIG. 13 are cross-sectional views of a MEMS-CMOS wafer fabricated at various stages, in accordance with some embodiments of the present disclosure. In FIG. 5, a MEMS wafer **500** is provided. The MEMS wafer **500** comprises a substrate **502**, an oxide layer **504**, and a semiconductor layer **506**. The substrate **502** has a first surface **5022** and a second surface **5024** opposite to the first surface **5022**. The oxide layer **504** is disposed on the first surface **5022** of the substrate **502**. The semiconductor layer **506** is disposed on the oxide layer **504**. In FIG. 5, a first scribe line region **508**, a first die region **510**, a second scribe line region **512**, a second die region **514**, and a wafer edge region **516** are defined in MEMS wafer **500**.

[0032] In FIG. 6, an etching MEMS wafer **600** is provided. According to some embodiments, the oxide layer **504** and the semiconductor layer **506** in the first scribe line region **508** are not etched. The oxide layer **504** and the semiconductor layer **506** in the second scribe line region **512** are etched. The oxide layer **504** and the semiconductor layer **506** in the wafer edge region **516** are etched. The oxide layer **504** in the first die region **510** and the second die region **514** are not etched. The semiconductor layer **506** in the first die region **510** and the second die region **514** are etched according to a pattern. The pattern corresponds to the top surfaces of the individual dies. After the etching process, an oxide layer **602** is disposed over the oxide layer **504** and the semiconductor layer **506**. However, the second scribe line region **512** and the wafer edge region **516** are not disposed by the oxide layer **602**. In other words, in the second scribe line region **512** and the wafer edge region **516**, a scribe line trench **604** and a wafer edge line trench **606** are formed to expose the first surface **5022** of the substrate **502**.

[0033] In FIG. 7, a structured MEMS wafer **700** is provided. According to some embodiments, the scribe line trench **604** and the wafer edge line trench **606** are disposed with a semiconductor material to form a scribe line contact **702** and a wafer edge line contact **704**, respectively. The scribe line contact **702** has a first end contacted with the first surface **5022** of the substrate **502** and a second end configured to be a contacting surface **708** of the structured MEMS wafer **700**. The wafer edge line contact **704** has a first end contacted with the first surface **5022** of the substrate **502** and a second end configured to be the contacting surface **708** of the structured MEMS wafer **700**. The semiconductor material may be polycrystalline silicon. In the first die region **510** and the second die region **514**, the predetermined mechanical structures of the dies are formed over the semiconductor layer **506** by using the semiconductor material. In the first scribe line region **508**, the oxide layer **706** is formed over the semiconductor layer **506**. The oxide layer **706** is also filled in the spaces among the mechanical structures of the dies. The contacting surface **708** of the structured MEMS wafer **700** may protrude from the oxide layer **706**. In addition, the heights of the scribe line contact **702** and a wafer edge line contact **704** are **H1**. The thickness of the substrate **502** is **H2'**. The width of the first scribe line region **508** is **W1**. According to some embodiments, the width **W1** is larger than the height **H1**. For example, the width **W1** may be two times larger than the height **H1**. According to some embodiments, the height **H1** may be about 10 μm , and the thickness **H2'** may be about 600 μm .

[0034] In FIG. 8, a bonding MEMS-CMOS wafer **800** is provided. According to some embodiments, a CMOS wafer **802** having integrated circuits formed therein is bonded to the contacting surface **708** of the structured MEMS wafer **700**. The contacting surface **708** of the structured MEMS wafer **700** is eutectic bonded to the predetermined interconnections **804** of the CMOS wafer **802** such that the structured MEMS wafer **700** may be electrically connected to the

CMOS wafer **802**. The bonding MEMS-CMOS wafer **800** has a bottom surface **806** and a top surface **808**. A release barrier **810** is disposed over the CMOS wafer **802**. The thickness of the CMOS wafer **802** is $H3'$. According to some embodiments, the thickness $H3'$ may be about 725 μm . Accordingly, the total thickness of the bonding MEMS-CMOS wafer **800** in FIG. **8** may be about 1335 μm .

[0035] In FIG. **9**, a first thinned down MEMS-CMOS wafer **900** is provided. According to some embodiments, a thinning process is performed upon the bottom surface **806** at the CMOS wafer **802** to form a thinned down CMOS wafer **902** with a bottom surface **904**. The thickness of the thinned down CMOS wafer **902** is $H3$. The thickness $H3$ of the thinned down CMOS wafer **902** may be about 240 μm . Accordingly, the total thickness of the first thinned down MEMS-CMOS wafer **900** in FIG. **9** may be about 850 μm .

[0036] In FIG. **10**, a second thinned down MEMS-CMOS wafer **1000** is provided. According to some embodiments, a thinning process is performed upon the top surface **808** at the substrate **502** to form a thinned down substrate **1002** with a thinned down surface **1004**. The thickness of the thinned down substrate **1002** is $H2$. The thickness $H2$ of the thinned down substrate **1002** may be about 290 μm . Accordingly, the total thickness of the second thinned down MEMS-CMOS wafer **1000** in FIG. **10** may be about 540 μm . According to some embodiments, the bottom surface **904** of the thinned down CMOS wafer **902** corresponding to the first die region **510** and the second die region **514** may be etched to have a first trench **1006** and a second trench **1008** respectively. The first trench **1006** and the second trench **1008** are arranged to expose the oxide layer **1010** in the thinned down CMOS wafer **902**.

[0037] In FIG. **11**, an etched MEMS-CMOS wafer **1100** is provided. According to some embodiments, an etching process is performed upon the thinned down surface **1004** at the thinned down substrate **1002**. The etching process is arranged to etch the thinned down substrate **1002** corresponding to the first die region **510** and the second die region **514** to form a first trench **1102** and a second trench **1104**, respectively. According to some embodiments, the first scribe line region **508**, the second scribe line region **512**, and the wafer edge region **516** are masked by a photoresist layer, and the first die region **510** and the second die region **514** are not masked by the photoresist layer. Then, the thinned down substrate **1002** corresponding to the masked region (i.e. the first scribe line region **508**, the second scribe line region **512**, and the wafer edge region **516**) are not etched, and the thinned down substrate **1002** corresponding to the unmasked region (i.e. the first die region **510** and the second die region **514**) are etched. After the etching process, the first trench **1102** and the second trench **1104** expose the oxide layer **504** in the second thinned down MEMS-CMOS wafer **1000**. The thinned down substrate **1002** corresponding to the first scribe line region **508**, the second scribe line region **512**, and the wafer edge region **516** form three etching substrates **1106**, **1108**, **1110** respectively. Accordingly, the etching substrate **1106** contacts the oxide layer **504**, the etching substrate **1108** contacts the scribe line contact **702**, and the etching substrate **1110** contacts the wafer edge line contact **704**. According to some embodiments, the widths of the etching substrates **1106**, **1108** are $W1$, the width of the etching substrate **1110** is $W3$, and the thickness of the oxide layer in the etched MEMS-CMOS wafer **1100** is $H1$. The width $W1$ of the etching substrate **1106** is at least two times larger than the thickness $H1$ of the oxide layer in the etched MEMS-CMOS wafer **1100**. For example, the thickness $H1$ may be about 10 μm , the width $W1$ may be about 60 μm , and the width $W3$ may be about 5 mm.

[0038] In FIG. **12**, a MEMS-CMOS wafer **1200** is provided. According to some embodiments, an etching process is performed upon the etched MEMS-CMOS wafer **1100** for removing the oxide layers **504**, **706**, **1010** in the etched MEMS-CMOS wafer **1100**. According to some embodiments, the width $W1$ of the oxide layer **504** sandwiched between the etching substrate **1106** and the semiconductor layer **506** is at least two times larger than the thickness $H1$ of the oxide layer in the etched MEMS-CMOS wafer **1100**. When the speed of etching performed upon the oxide layer (i.e. **706**, **1100**) in the etched MEMS-CMOS wafer **1100** is substantially equal to the speed of etching

performed upon the oxide layer **504**, a portion of the oxide layer **504** sandwiched between the etching substrate **1106** and the semiconductor layer **506** may remain when the oxide layer (i.e. **706**, **1100**) in the etched MEMS-CMOS wafer **1100** is etched. Accordingly, in the first scribe line region **508**, the outer portion **1202** of oxide layer **504** sandwiched between the etching substrate **1106** and the semiconductor layer **506** is etched or released by the etching process, and the inner portion **1204** of the oxide layer **504** sandwiched between the etching substrate **1106** and the semiconductor layer **506** is not etched by the etching process. Therefore, the inner portion **1204** may be used to fix the etching substrate **1106** on the semiconductor layer **506**. According to some embodiments, to firmly fix the etching substrate **1106** on the semiconductor layer **506** by the inner portion **1204**, the width **W1** of the oxide layer **504** sandwiched between the etching substrate **1106** and the semiconductor layer **506** may be three times larger than the thickness **H1** of the oxide layer in the etched MEMS-CMOS wafer **1100**. In the second scribe line region **512** and the wafer edge region **516**, the etching process are stopped by the scribe line contact **702** and the wafer edge line contact **704**, respectively. Accordingly, the etching substrates **1106**, **1108**, and **1110** protruding from the surface **1206** are formed. The etching substrates **1106** and **1108** may be regarded as the first scribe line **218** and the second scribe line **220** in FIG. 2, and the etching substrate **1110** may be regarded as the wafer edge line **108** in FIG. 3. As the etching substrates **1106**, **1108**, and **1110** increase the effective thickness **H5** (e.g. 540 μm) of the MEMS-CMOS wafer **1200**, the wafer tension of the MEMS-CMOS wafer **1200** is increased such that the warpage or deformation of the MEMS-CMOS wafer **1200** can be alleviated when the handling system handles the MEMS-CMOS wafer **1200** during transportation.

[0039] In FIG. 13, a dicing MEMS-CMOS wafer **1300** is provided. According to some embodiments, the MEMS-CMOS wafer **1200** is disposed on a dicing tape **1302**. Then, the etching substrates **1106**, **1108**, and **1110** are cut by a cutting tool to singulate a first die **1304** and **1306**. After dicing, the etching substrates **1106**, **1108**, and **1110** with the thickness **H5** are removed, and the dies **1304** and **1306** with relatively small thickness **H4** are singulated.

[0040] Therefore, by using the proposed method, the effective thickness **H5** of the MEMS-CMOS wafer **1200** is increased, the array of dies **1304** and **1306** can be designed to have relatively small thickness **H4** while the handling system still can handle the MEMS-CMOS wafer **1200** without being delaminated.

[0041] In some embodiments of the present disclosure, a method of fabricating a semiconductor structure is disclosed. The method comprises: providing a first wafer; providing a second wafer having a first surface and a second surface opposite to the first surface; contacting the first surface of the second wafer with the first wafer; and forming a plurality of scribe lines on the second surface of the second wafer, wherein the formation of the plurality of scribe lines includes removing portions of the second wafer from the second surface towards the first surface to form a third surface between the first surface and the second surface, and the plurality of scribe lines protrudes from the third surface of the second wafer.

[0042] In some embodiments of the present disclosure, a method of fabricating a semiconductor structure is disclosed. The method comprises: providing a first wafer; providing a second wafer having a first surface, a second surface opposite to the first surface, a first semiconductor layer, an oxide layer disposed over the first semiconductor layer and a second semiconductor layer disposed over the oxide layer, wherein the first semiconductor layer, the oxide layer and the second semiconductor layer are disposed between the first surface and the second surface; contacting the first surface of the second wafer with the first wafer; removing portions of the second semiconductor layer to form a plurality of scribe lines; and removing portions of the oxide layer exposed from the plurality of scribe lines.

[0043] In some embodiments of the present disclosure, a method of fabricating a semiconductor structure is disclosed. The method comprises: providing a first wafer; providing a second wafer having a first surface, a second surface opposite to the first surface, a first semiconductor layer, an

oxide layer disposed over the first semiconductor layer and a second semiconductor layer disposed over the oxide layer, wherein the first semiconductor layer, the oxide layer and the second semiconductor layer are disposed between the first surface and the second surface; contacting the first surface of the second wafer with the first wafer; removing portions of the second semiconductor layer to form a plurality of scribe lines and a plurality of wafer edge lines; and cutting from the second surface of the second wafer towards the first wafer and along the plurality of scribe lines and the plurality of wafer edge lines.

[0044] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a first wafer; and a second wafer, having a first surface attached to the first wafer and a second surface facing away the first wafer and including: a first semiconductor layer adjacent to the second surface; a second semiconductor layer disposed adjacent to the first surface; and an oxide layer between the first semiconductor layer and the second semiconductor layer, wherein a plurality of scribe lines are included in the first semiconductor layer.
2. The semiconductor structure of claim 1, wherein the first wafer is a complementary metal oxide semiconductor (CMOS) wafer, and the second wafer is a micro-electro mechanical system (MEMS) wafer.
3. The semiconductor structure of claim 1, wherein the first wafer is a micro-electro mechanical system (MEMS) wafer, and the second wafer is a complementary metal oxide semiconductor (CMOS) wafer.
4. The semiconductor structure of claim 1, wherein a wafer edge line is included at an edge of the second wafer.
5. The semiconductor structure of claim 4, wherein the wafer edge line surrounds the plurality of scribe lines.
6. The semiconductor structure of claim 1, wherein the oxide layer and the second semiconductor layer are a plurality of scribe line contacts in contact with the plurality of scribe lines.
7. The semiconductor structure of claim 1, wherein the first semiconductor layer is a wafer edge line, the oxide layer and the second semiconductor layer are a wafer edge line contact, and the wafer edge line contact is in contact with the wafer edge line.
8. A semiconductor structure, comprising: a first substrate; and a second substrate, attached to the first substrate and having a first surface facing away the first substrate and including: a first semiconductor layer; an oxide layer, disposed on the first semiconductor layer; and a second semiconductor layer, disposed on the oxide layer, wherein a plurality of scribe lines are included in the first semiconductor layer, and a wafer edge line is at an edge of the second substrate and surrounds the plurality of scribe lines.
9. The semiconductor structure of claim 8, wherein the plurality of scribe lines are a plurality of intersecting scribe lines.
10. The semiconductor structure of claim 8, wherein the wafer edge line and the plurality of scribe lines have a same thickness.
11. The semiconductor structure of claim 8, wherein the first substrate is a complementary metal

- oxide semiconductor (CMOS) wafer, and the second substrate is a micro-electro mechanical system (MEMS) wafer.
- 12.** The semiconductor structure of claim 8, wherein the first substrate is a micro-electro mechanical system (MEMS) wafer, and the second substrate is a complementary metal oxide semiconductor (CMOS) wafer.
- 13.** The semiconductor structure of claim 8, wherein a width of the wafer edge line is greater than a width of a scribe line.
- 14.** The semiconductor structure of claim 8, wherein the oxide layer and the second semiconductor layer are configured to be a wafer edge line contact, and the wafer edge line contact is in contact with the wafer edge line.
- 15.** A semiconductor structure, comprising: a CMOS wafer; and a MEMS wafer, attached to the CMOS wafer and including: a first semiconductor layer; an oxide layer, disposed on the first semiconductor layer; and a second semiconductor layer, disposed on the oxide layer, wherein a plurality of scribe lines are included in the first semiconductor layer.
- 16.** The semiconductor structure of claim 15, wherein the wafer edge line and the plurality of scribe lines have a same thickness.
- 17.** The semiconductor structure of claim 15, wherein a width of the wafer edge line is greater than a width of each one of the plurality of scribe lines.
- 18.** The semiconductor structure of claim 15, further comprising: a plurality of scribe line regions and a plurality of die regions over the second semiconductor layer, wherein a scribe line contact is disposed within one of the plurality of scribe line regions.
- 19.** The semiconductor structure of claim 15, wherein the second surface is configured to have a wafer edge line on an edge of the MEMS wafer.
- 20.** The semiconductor structure of claim 19, wherein the width of each one of the plurality of scribe lines is greater than a thickness between the first surface and the fifth surface.
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